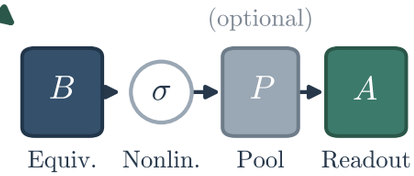
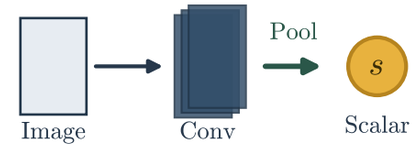
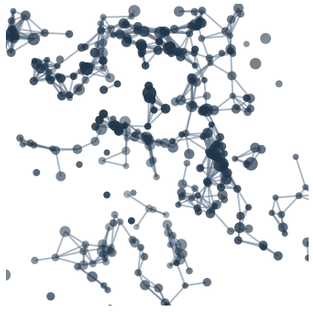
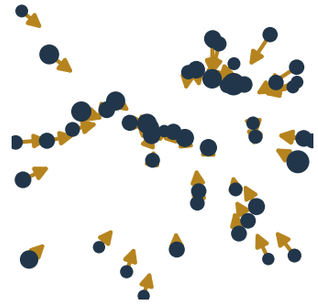
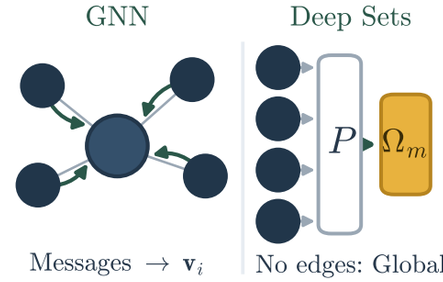
THE THREE GEOMETRIES OF A LEARNING PROBLEM

	1. DOMAIN	2. SIGNAL	3. MODEL
	Where the data live, Ω	What lives on it	What we build
GENERAL	 Grid Sphere Graph	Scalar Vector Tensor  $s \mapsto s$ $\mathbf{v} \mapsto R\mathbf{v}$ $\mathbf{T} \mapsto R\mathbf{T}R^\top$ Invariant Equivariant Equivariant	Implies  (optional) B σ P A Equiv. Nonlin. Pool Readout $f = A \circ (P) \circ \sigma \circ B \circ \cdots \circ B$
	Grid, sphere, point cloud, graph Translation, rotation, permutation	A field on Ω, with its symmetry (Here R is a rotation)	The geometric blueprint Its symmetry $\subseteq$ the domain's
	Image on a grid	Example inference task: Time since last merger	Implies CNN
GALAXY			 Image Conv Pool Scalar
	Translations $T(2)$, 90° rotations (D_4)	Rotate by any θ: Same answer $\Rightarrow$ invariant under $SO(2)$	Equivariant convs, then global pool to a scalar
	Halo point cloud	Example inference task: Velocity vectors, or ΛCDM	Implies GNN / Deep Sets
HALOS			 GNN Deep Sets P Ω_m Messages $\rightarrow \mathbf{v}_i$ No edges: Global
	Permutation Σ_n; Linking length $\Rightarrow$ locality	Vector: $O(3)$-**equivariant** ΛCDM: $E(3)$-**invariant**	Use GNN for local interactions; Deep Sets for global param invariance.

Left to right: The domain and the signal imply the model.